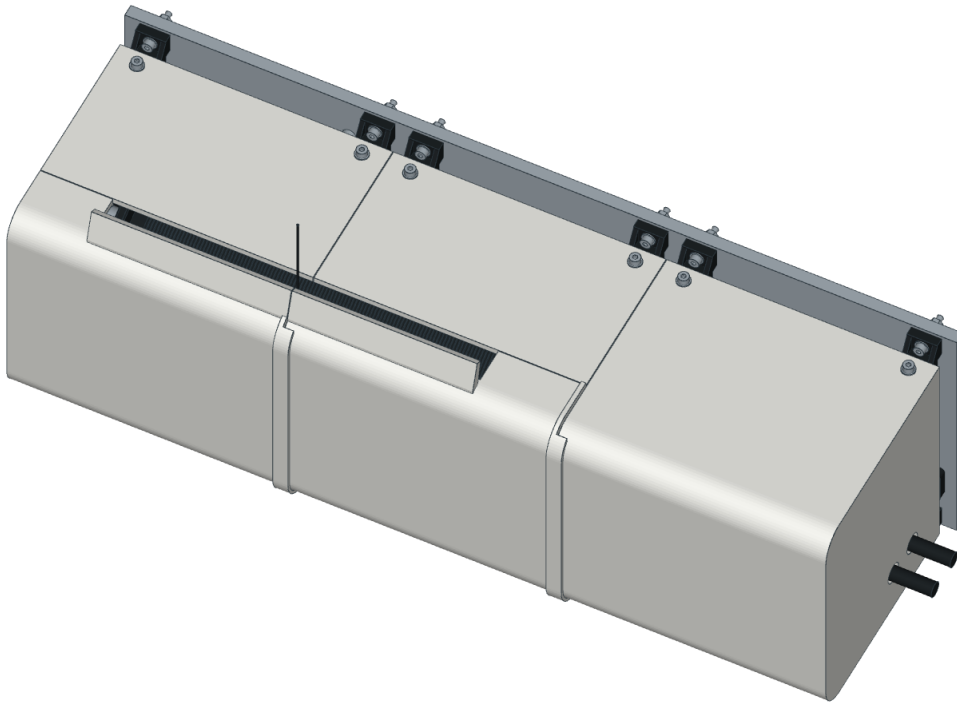


Build the actual parts

Passive-line winch: two drum sections, 340 mm shaft, 550 x 180 mm base. The passive-line winch still has a motor.



REAL GEOMETRY, STEP BY STEP

Every 3D illustration is rendered from the supplied STL meshes. ARBI prints are exported unchanged from the source CAD; bought hardware is represented by clearly labelled reference models.

What is corrected

Grooved hollow drum sections and their three ribs; flat handed flanges; one alignment pin at each joint; the actual split clamp, offset bearing supports and motor stand.

Build unpowered first. This booklet covers assembly and hand-spin checks. CAD, desk feet and full cover remain concept-unvalidated. Cover installation follows on pages 15-18; loaded tests require a separate checked setup.

01 Printed parts

For one passive winch. Each picture is the actual STL; pictures use independent scales. Print/export file names are listed in the model pack.



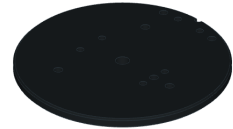
P01 Passive section 1

1 x / one recessed dot



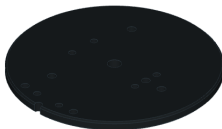
P02 Passive section 2

1 x / two recessed dots



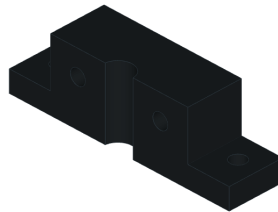
P03 Left flange

1 x



P04 Right flange

1 x / handed part



P05 Shaft clamp half

2 x / identical prints



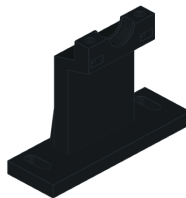
P06 Line tail clamp

1 x / two underside channels



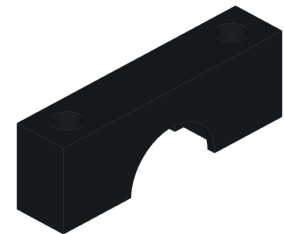
P07 Alignment pin

3 x / one per joint



P08 Bearing lower

2 x / feet face outward



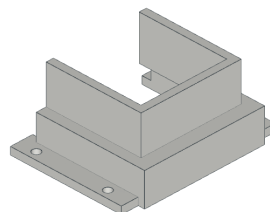
P09 Bearing cap

2 x



P10 Motor stand

1 x



P11 Coupling cover

1 x / revision 0.1.1

Desk feet are optional

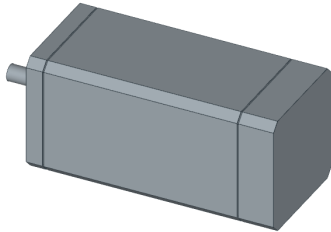
Two short + two long.
See page 13.

Powered drum sections are in the ZIP, but are not used here.

Deburr holes, locating sockets and mating faces. Dry-fit the shaft and bearings. The drum groove must stay continuous across the middle joint.

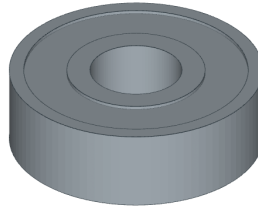
02 Bought parts

Reference STLs describe the hardware for these illustrations. They are not printed replacements for bearings, shafts, fasteners, the motor or coupling.



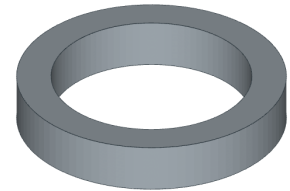
H01 Motor

1 x / 23HS40 reference



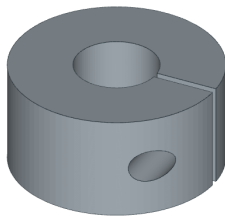
H02 608 bearing

2 x / 8 x 22 x 7 mm



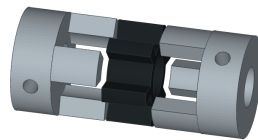
H03 Inner-ring spacer

2 x / 8.2 ID, 11 OD, 2 thick



H04 Shaft collar

2 x / 8 mm bore



H05 Jaw coupling

1 set / two hubs + spider



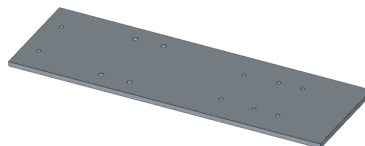
H06 Drum shaft

1 x / dia 8, nominal 340 mm



H07 Drum tie rods

3 x / M5, nominal 280 mm



H08 Aluminium plate

1 x / 550 x 180 x 8 mm



H09 Fasteners

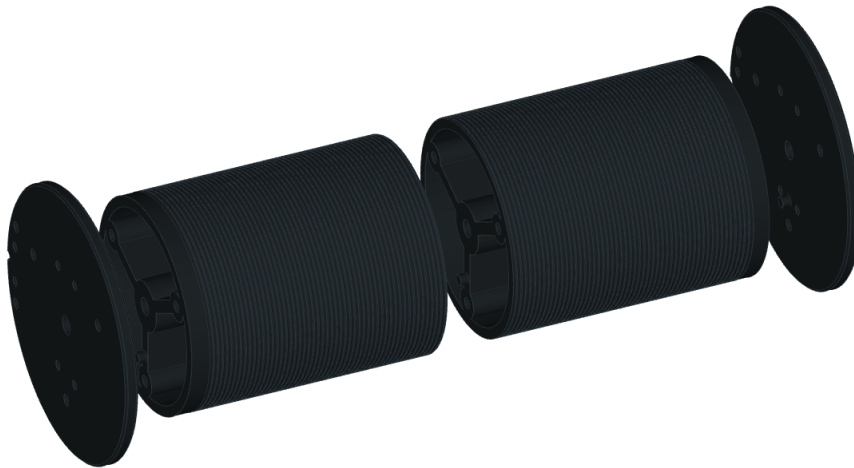
See complete table, page 14

The motor body/shaft envelope follows the supplier specification. Mounting pitch and pilot, collar outline and coupling internals are nominal. Match the received hardware before tightening.

Keep steel spacers on the bearing **inner rings**. Ordinary large washers may rub bearing seals or the housing. M5 nuts inside the bearing lowers must be plain nuts, not tall nylocs.

03 Key the drum sections

Do not turn either body section over. The right flange is a separate handed print, turned over so its locating socket faces inward.



LEFT FLANGE



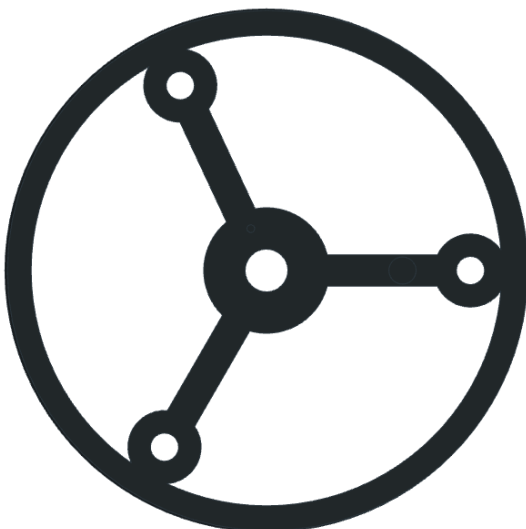
SECTION 1



SECTION 2



RIGHT FLANGE



1 Find the dot marks

Section 1 has one recessed dot; section 2 has two. Keep the body sections in their original print orientation.

2 Use one pin per joint

Three pins total: left flange / section 1, section 1 / section 2, section 2 / right flange.

The tie-rod pattern is asymmetric: 0, 115 and 240 degrees. All three rod holes and ribs must align without forcing. Do not try to rotate a section by 120 degrees.

04 Close and tie the drum

Keep the end faces flush. Tighten the three tie rods progressively, alternating between them.

1 Preload four clamp-foot screws

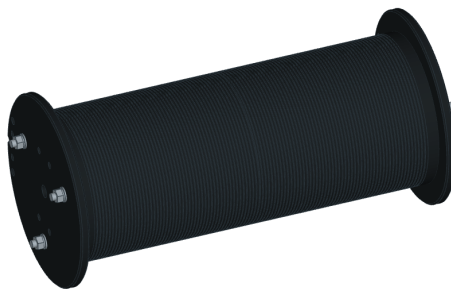
Before closing the drum, feed four M4 x 25 screws and washers through the right flange from its inside face. Their shafts point outward toward the split clamp.



4 x M4 x 25 + 4 washers, inserted before closing

2 Fit the tie rods

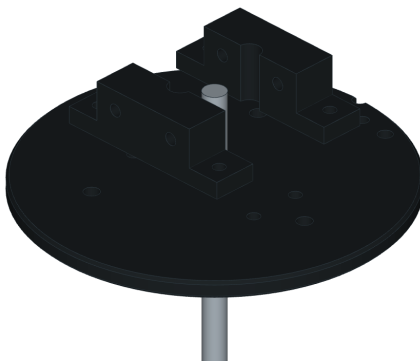
Pass three M5 x 280 rods through the complete stack. Put one M5 washer and one M5 locknut at each end: six washers and six nuts total.



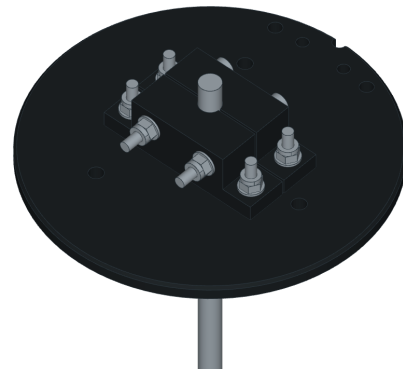
Faces should close flush without crushing the prints. No validated tightening torque is specified. Recheck the groove seam and all three joints before fitting the shaft clamp.

05 Clamp the shaft

The clamp is two identical rectangular halves. Rotate the second half 180 degrees around the shaft; do not substitute a single collar.



BRING THE TWO HALVES TOGETHER



CLAMP FIRST, FEET SECOND

1 Slide in the 8 mm shaft

Pass the shaft through the drum and leave it movable while fitting the two clamp halves onto the four preloaded flange screws.

2 Tighten the cross-bolts first

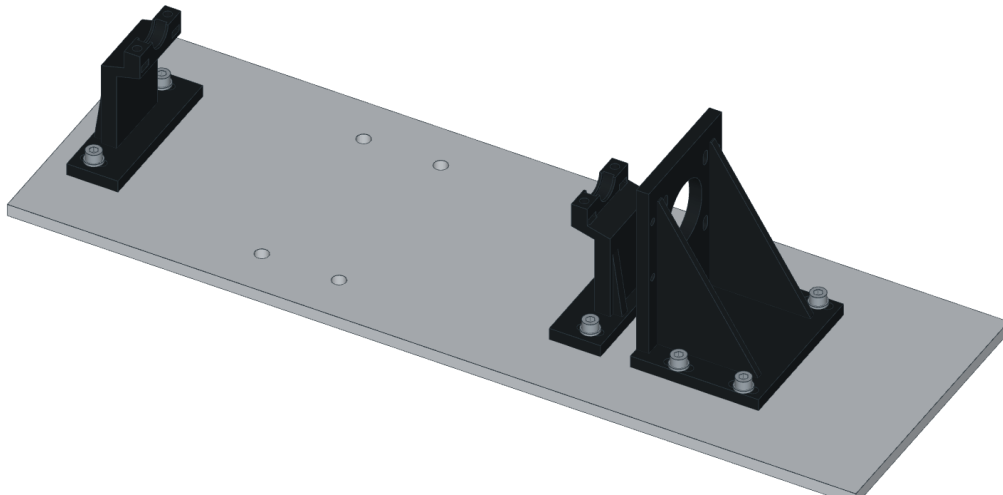
Use two M4 x 45 bolts, four washers and two metal locknuts. Tighten evenly to grip the shaft. Keep the designed split gap; do not force the halves fully shut.

Then secure the four clamp feet with four additional M4 washers and four locknuts. The shaft continues out of view in these close-ups.

For the 340 mm shaft, start with about 27 mm projecting beyond the non-motor flange; verify final coupling engagement on page 10. Fit the tail clamp on page 11.

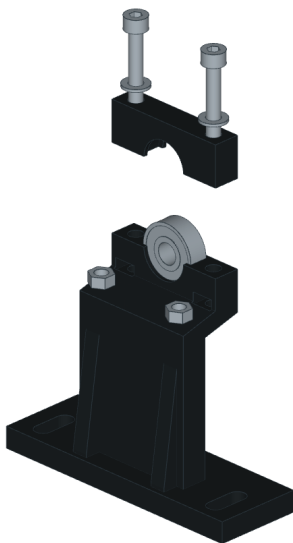
06 Prepare the base supports

Use the drilling template at 100% / actual size. The long feet of both bearing lowers must point outward, away from the drum.



8 x M6 x 30 / 16 washers / 8 locknuts

Attach both bearing lowers and the motor stand loosely. Their slots provide alignment travel; keep them free to move until the shaft rotates freely.



1 Insert the plain nuts

Slide two plain M5 nuts into the side windows of each bearing lower before fitting its cap.

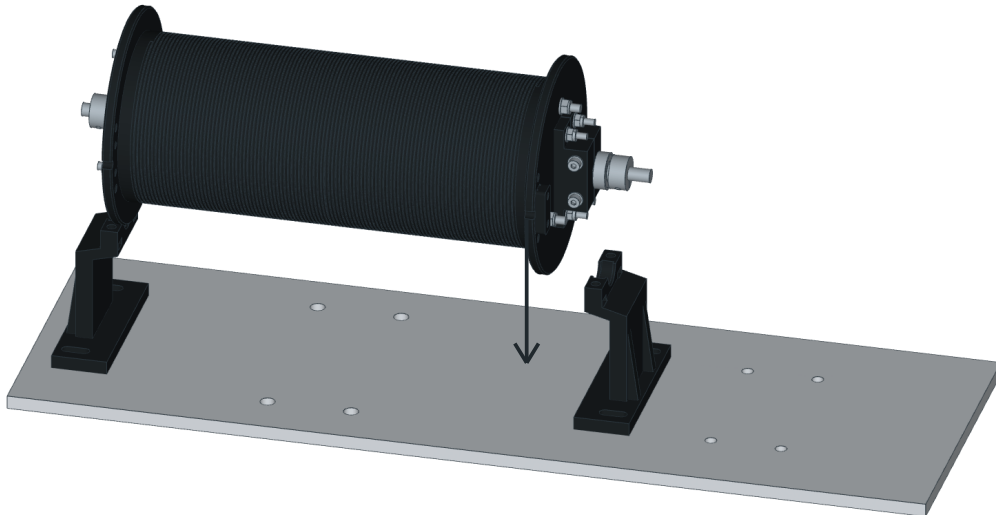
2 Keep caps loose

Each cap uses two M5 x 35 screws and two washers. Add the bearing/shaft assembly first, then the caps on page 8.

The right lower is the same print rotated 180 degrees around the vertical axis.

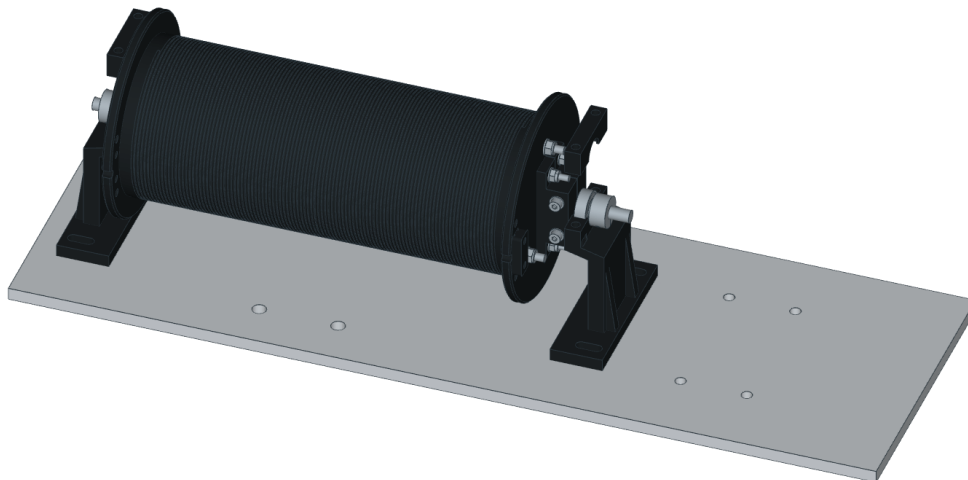
07 Place the shaft assembly

Slide both bearings, steel spacers and collars onto the shaft before closing the bearing caps. Keep collars loose initially.



LOWER BOTH BEARINGS INTO THEIR OPEN SEATS

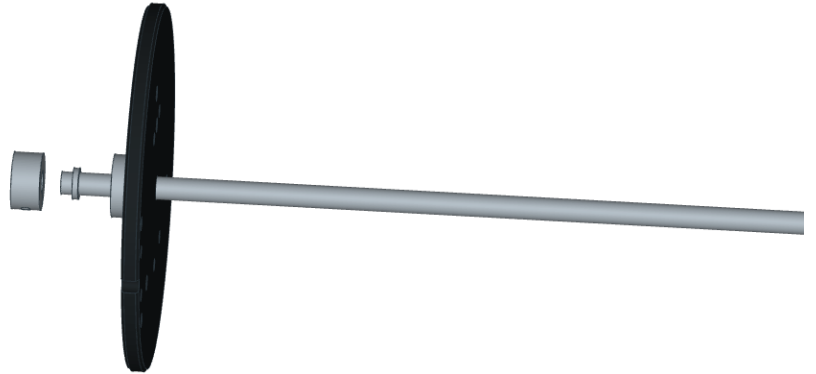
Place the complete drum and shaft on the two bearing lowers. The bearings sit in the circular seats; the drum flanges and clamp stay clear of both supports.



Fit both caps using four M5 x 35 screws and four washers into the previously inserted plain nuts. Tighten gently and evenly. The 0.4 mm design split is not a reason to crush the housing shut.

08 Set collars and alignment

Spacers and collars sit outside the two bearings, away from the drum. Leave slight axial freedom; do not preload the bearings.



NON-MOTOR END: collar > spacer > bearing > flange

The thin spacer must contact only the bearing inner ring. Keep its outer edge clear of the bearing seal and printed housing. The collar locks to the shaft; it must not clamp the housing.

MOTOR END

Drum / split shaft clamp > bearing > 2 mm spacer > shaft collar > coupling.

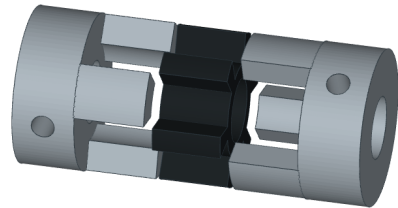
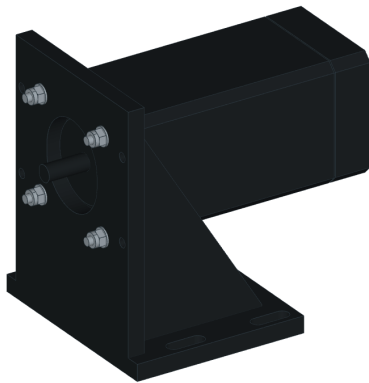
1 Align the bearing lowers

Turn the shaft by hand while adjusting the M6 slots. Tighten base bolts progressively and check rotation again after each adjustment.

Stop if you feel scraping, binding or a notchy bearing. Reposition the support or collar before continuing.

09 Align motor and coupling

Install the motor with its shaft toward the drum. The coupling connects aligned shafts; do not use it to pull misaligned mounts together.

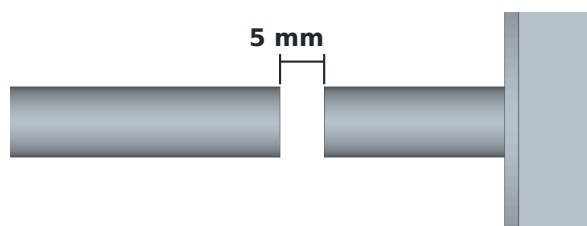
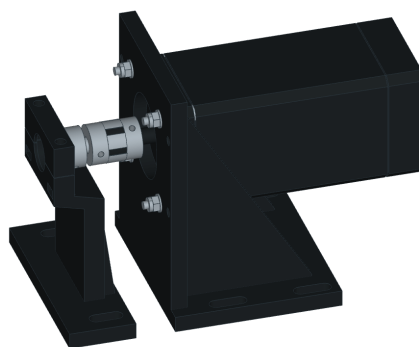


Motor mounting

Fit four fasteners matched to the received motor flange. M4 x 25 is only the repository allowance; verify hole type and engagement.

Coupling

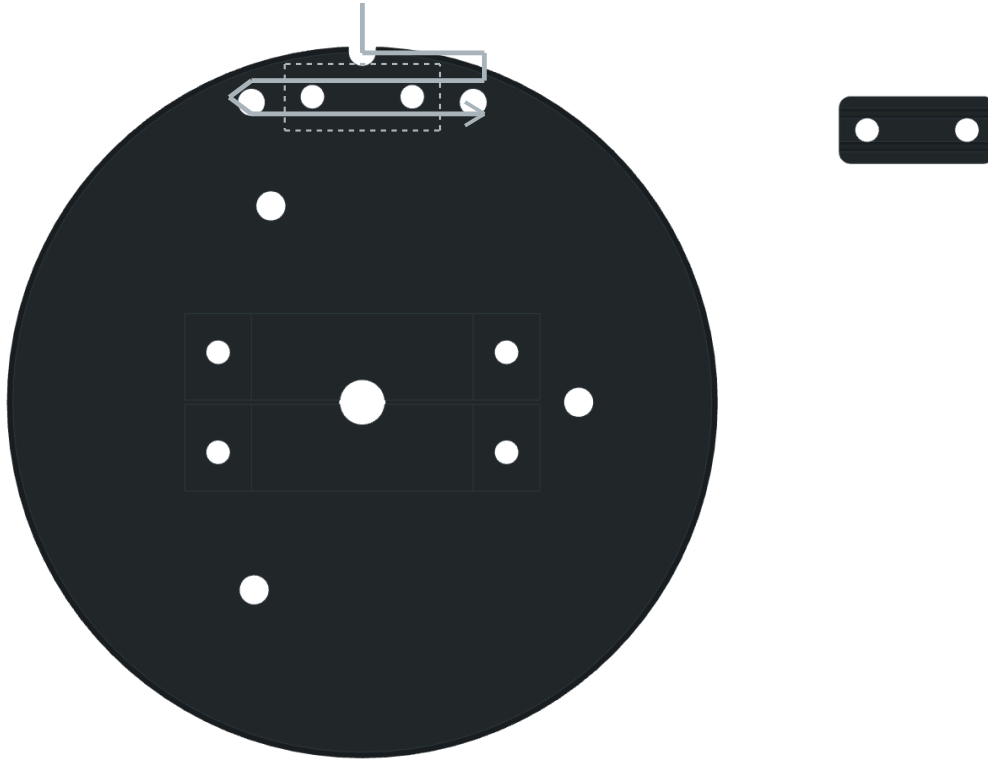
Place one hub on each shaft and the elastomer spider between them. Seat and lock the hubs according to the actual coupling.



Nominal CAD setup: about 10 mm shaft engagement at each coupling end, leaving 5 mm between shaft tips. Verify actual hub/spider geometry. **The shaft tips must not touch.**

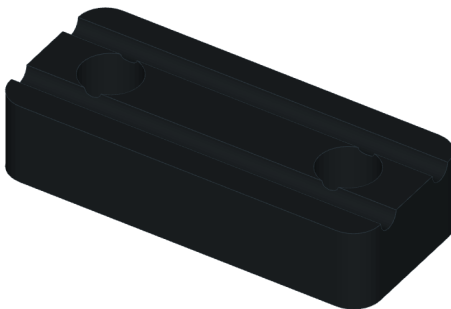
10 Secure the line tail

Fit the tail clamp on the outside of the right flange. It grips two bare Dyneema legs in its underside channels.



Route over the end margin and through the rim notch, then double back through both clamp channels. The silver overlay shows the two legs and return loop; the clamp is moved aside to show its underside.

Use two M4 x 25 screws, four washers and two locknuts. Smooth the notch and channels. Keep at least **three reserve wraps** on the drum at maximum payout.

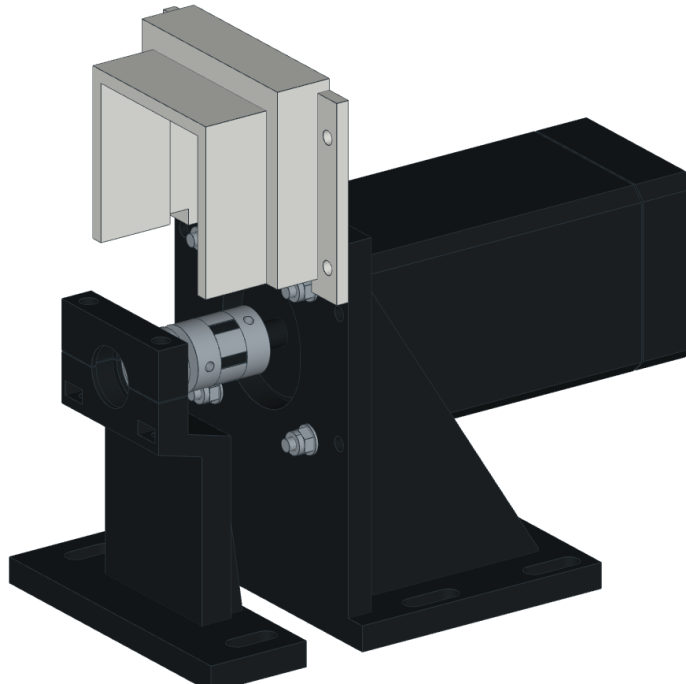


The two shallow channels on the clamp underside carry the outgoing and returning bare line. Keep the line clear of screw holes.

Finish the line termination and confirm free hand rotation before fitting the revised coupling cover on the next page.

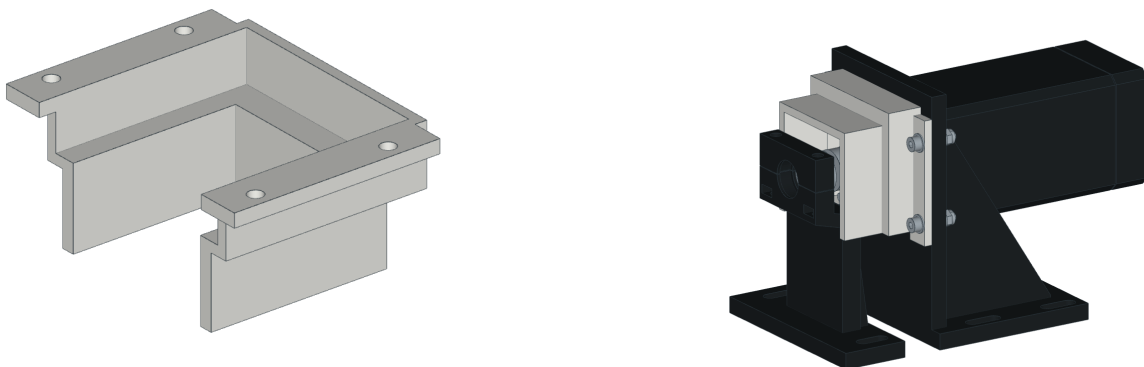
11 Fit the revised coupling cover

Reprint P11 only: winch-coupling-guard-r0.1.1.stl. The existing motor stand and its four cover mounting holes remain compatible.



WIDER MOTOR END CLEARS THE FOUR FASTENER STACKS

Tighten and align the motor first. Lower the cover over the coupling with its open underside downward; the widened end faces the motor stand. The drawing includes representative motor screws, washers and nuts.

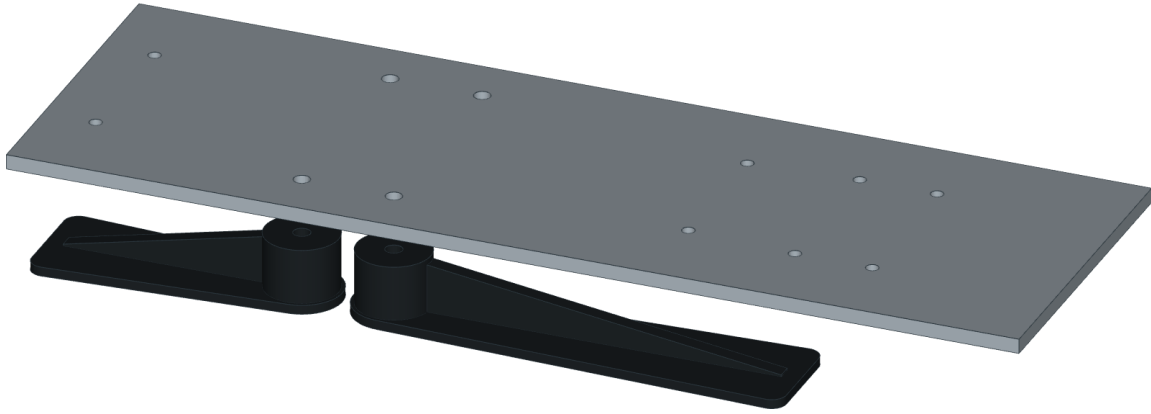


Check your hardware: maximum 12 mm across each motor fastener stack and 14 mm projection from the coupling-side stand face, including screw tips. The recess adds 1 mm clearance and allows the full ± 2 mm vertical slot adjustment.

Use four M4 x 20 cover screws, eight 9 mm OD washers and four locknuts. Seat the flange without force. The revised print still needs a physical fit check.

12 Desk feet and final checks

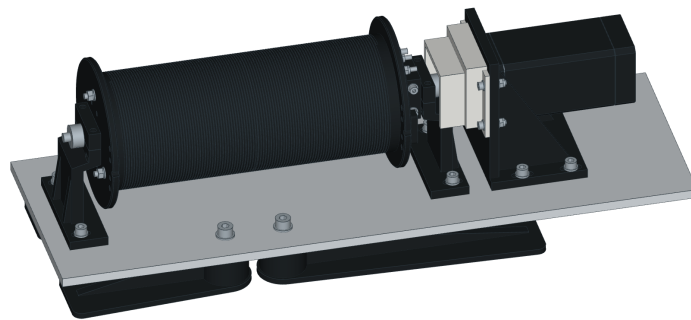
Optional desk setup for the passive base. The four existing post holes take the feet; there are no new holes to drill.



2 SHORT FEET: away from motor

2 LONG FEET: toward motor

Use four M8 x 35 bolts, eight standard washers and four M8 locknuts on the 8 mm plate. Recess the nuts inside the feet. Plate underside is 35 mm above the desk; existing screws may protrude at most 25 mm.



- ☐ All drum joints close flush; the groove is continuous.
- ☐ Shaft turns freely; collars do not preload bearings.
- ☐ Rods, nuts, flanges and clamp clear the supports through a full turn.
- ☐ Motor shafts align; coupling has axial clearance; all feet sit flat.

Complete these checks unpowered. Desk feet are not anchors for loaded cable tests.

13 Fasteners and model notes

Quantities below are for one passive winch. Nuts and washers are additional to the bolts or rods listed.

Location	Bolts / rods	Washers	Nuts
Drum tie rods	3 x M5 x 280	6 x M5	6 x M5 lock
Clamp feet to flange	4 x M4 x 25	8 x M4	4 x M4 lock
Clamp cross-bolts	2 x M4 x 45	4 x M4	2 x M4 lock
Line tail clamp	2 x M4 x 25	4 x M4	2 x M4 lock
Bearing caps	4 x M5 x 35	4 x M5	4 x M5 plain
Base supports + stand	8 x M6 x 30	16 x M6	8 x M6 lock
Coupling cover	4 x M4 x 20	8 x M4	4 x M4 lock
Motor to stand	4, match actual motor	As required	As required
Optional desk feet	4 x M8 x 35	8 x M8	4 x M8 lock

MODEL PACK: 62 STL FILES

27 ARBI fabrication meshes are exported from the bundled source CAD. The coupling cover is **revision 0.1.1**; the other fabrication models remain 0.1.0. Includes drum parts, mounts, coupling guard, desk feet and eleven additional full-cover model IDs; use the quantities on page 15.

35 reference meshes cover bought hardware and the metal base plates. Their simplified threads, bearing internals, motor mounting details and coupling teeth are illustrative. They are not supplier-certified models or manufacturing replacements.

The source, STL hashes, dimensions and a manifest of every illustrated component are included. All exported meshes were checked for closed surfaces, consistent winding and positive volume. Geometry checks do not establish print fit or load capacity.

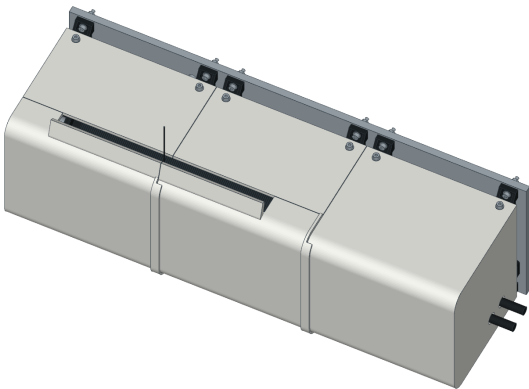
Source references

ARBI: github.com/gredice/arbi. Exact source files and SHA-256 hashes are included; see source-provenance.json.
Motor: StepperOnline kit 4-CLYS30-V20, model 23HS40-5004D-E1000. Supplier body/shaft dimensions checked 27 Sep 2026; supplier STEP download was unavailable.

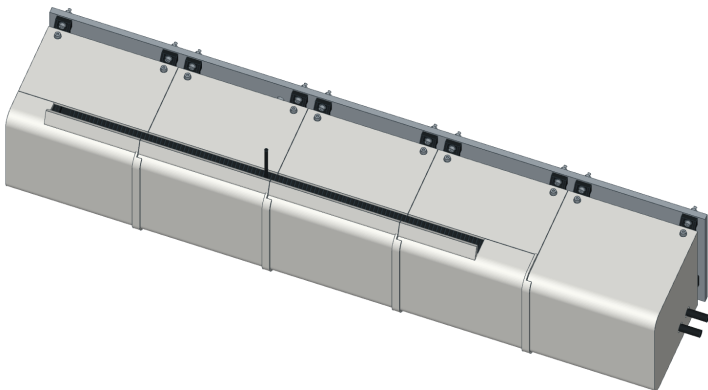
Powered-model STLs are included for completeness. This booklet assembles the passive-line version only; the longer powered base/shaft needs separate validation.

14 Full assembly cover

Added r0.1.0 kit under ADR-0007. Reuse the drivetrain and r0.1.1 coupling guard. Drill the extra base holes before fitting clips.



PASSIVE: 3 MAIN PANELS + 2 PAYOUT SHUTTERS



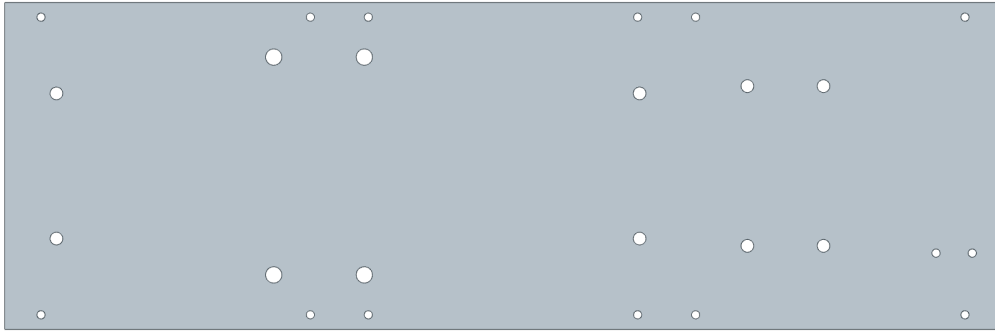
POWERED: 5 MAIN PANELS + 4 PAYOUT SHUTTERS

Added print	Passive	Powered
Left / middle / right main panels	1 / 1 / 1	1 / 2 / 1
Distinct powered transition, index 3	0	1
Identical variant payout shutters	2	4
Identical dark base clips	12	20
Fixed-loom base anchor	1	1

Main panels print on an axial seam; shutters on their outer face; clips and anchor on their base feet. Powered index 3 needs its own transition print: the slot stops at X=580.3 mm. Never use a third middle. All meshes start at print Z=0 and fit inside 240 mm per axis; inspect supports.

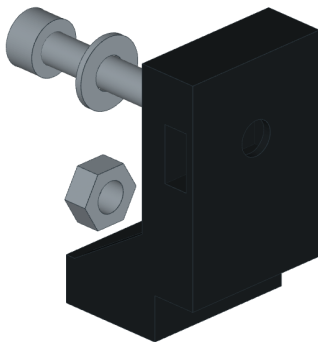
15 Clips, holes and fasteners

New holes only: keep all original bearing, motor-stand and post patterns. Base blanks stay 550 / 880 x 180 x 8 mm.



Clip holes, diameter 4.5 mm: $Y = \pm 82$. For each main panel i , $X = -46 + iP + 16$ and $X = -46 + (i+1)P - 16$. $P = 180.667$ mm passive ($i=0..2$), or 174.4 mm powered ($i=0..4$).

Anchor holes: $X = \text{cover end} - 22 \pm 10$, $Y = -48$. Cover end = 496 / 826 mm. Total: 14 / 22 added holes. The older drilling template omits these.



Each clip

M4 x 25 + two washers + locknut to base.

M4 x 16 + washer to plain M4 nut slid in from the side.

Use 9 mm OD washers and reviewed removable locking.

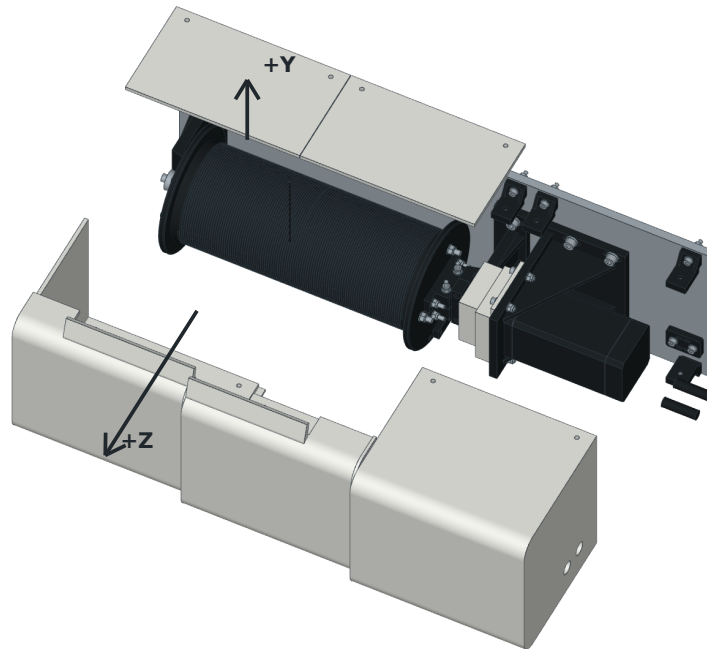
Inspect supports under the raised stem and nut pocket.

Per passive / powered kit: 12 / 20 shell screws, plain nuts and washers; 14 / 22 base+anchor screws and locknuts; 28 / 44 base washers. Anchor uses two soft loom wraps. Buy separately from original winch fasteners.

Slide captive nuts in before bolting clips down. Deburr and inspect the added holes. Install +Y upward on the post; Z points outward. Fit fixed looms before the panels.

16 Fit and remove the shell

Isolate power and mechanically secure/de-tension the line. Shutters make a threaded line removable from the shell without disconnecting it.



1 Fit right to left

Fit the right main panel, then middle panels from right to left, then left. The 7 mm exterior cuffs overlap the next panel. Fit payout shutters last; four clip screws total per panel position.

2 Remove shutters first

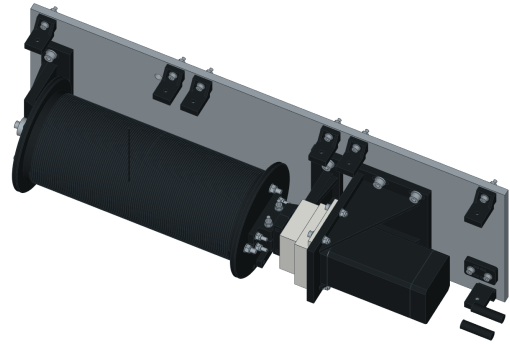
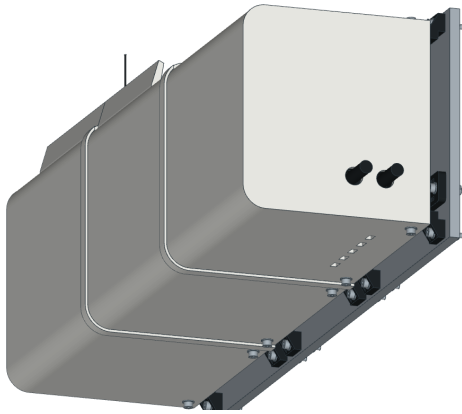
Remove +Y-side screws on the payout segments. Withdraw each lower shutter toward +Y at least 20 mm; the 24 mm payout slot opens down to the base face. Secure the line inside its corridor.

3 Lift left to right

Remove remaining shell screws. Lift main panels outward along +Z, left to right, at least 130 mm. Fixed clips and loom anchor stay on the base. Remove the original coupling guard separately.

17 Routing and acceptance

Powered base/shaft remain separate load cases. The full cover is concept-unvalidated: no physical fit, heat, weather or contact-protection acceptance.



Line: bidirectional payout along +Y across the full winding width. Slot Z=120..144 mm; represented corridor Z=124..140 mm. Nominal line diameters 1.5 / 4.5 mm. Smooth edges, traverse full travel and check fleet angle. The shell is not a fairlead.

Fixed looms: two 14 mm right-end ports at Y=-48, Z=50/75, for provisional ≤ 10 mm OD motor/encoder looms. Add soft edging, two wraps on the independent base anchor, measured bend radius and a drip loop. Connectors may require removing the right panel.

Powered interface still open: a nominal 30 x 18 x 18 mm slip-ring body bay is checked at X=W+54..W+84, Y=-71..-53, Z=71..89. No supplier geometry, bracket or rotating conductor loop is established. Do not power the hybrid line until these are designed and accepted.

Rain and heat: +Y brows and exterior cuffs; 14 mm base-face drain/air gap; five 10 x 5 mm downward motor vents. Inspect rain, condensation, UV, motor temperature and creep at actual duty. PLA is a fit prototype. The near-motor driver is not modeled here; its protected mounting/enclosure still needs electrical design.

Service: with shells off, existing cap/base/motor fasteners retain their access. Release the coupling and caps before lifting the drum/shaft. Repeat alignment, line routing and hand-spin checks on reassembly. Payout openings and skirt gaps can admit fingers/tools; safe guarding remains unverified.